# Adarsh Sahu

**Data Science | Applied Machine Learning | Quantitative Modeling**

Jamshedpur, India | adarshprivate678@gmail.com | +91 92040 88891 | portfolio | LinkedIn | GitHub

## Summary

Undergraduate at NIT Jamshedpur specializing in applied machine learning, probabilistic modeling, and statistical analytics. Proven track record developing dual-propensity classification systems, selective prediction models with calibrated abstention, and credit risk pipelines with strict data-leakage auditing and TreeSHAP explainability. Experienced in formulating constrained optimization frameworks, translating complex statistical outputs into measurable business ROI.

## Technical Skills

**Languages & Core Stack:** Python (NumPy, pandas, SciPy), SQL (PostgreSQL), C++, Git, Bash/Linux
**Machine Learning & Modeling:** Supervised/Unsupervised Learning, Gradient Boosted Trees (XGBoost, LightGBM), Classification, Learning-to-Rank (LTR), Time-Series (Prophet, ARIMA), TreeSHAP, Feature Selection, Imbalanced Learning (SMOTE)
**Statistical Methods & Evaluation:** A/B Testing, Hypothesis Testing, ROC-AUC, Gini Coefficient, KS Statistic, Precision-Recall, Calibration Curves (Brier Score), Spatial & Temporal Holdout Cross-Validation, Data Leakage Auditing
**AI Systems & Engineering:** FastAPI, FAISS (Vector Search), Embeddings, RAG Architectures, Docker, Streamlit, REST API Serving
**Analytics & BI Tools:** PostgreSQL (Advanced CTEs, Window Functions), Advanced Excel (Solver, What-If), Tableau, EDA

## Work Experience

**Data Analytics & Growth Intern** Jul 2026 – Present
*OA Helper* / *oahelper.in* *Remote — Ranchi, Jharkhand*

- Spearheaded analytics and user acquisition modeling across 50+ campus engineering clusters, tracking onboarding funnels and retention dynamics to guide product growth.

## Featured Machine Learning Projects

**Collections Intelligence — Dual-Propensity Credit Recovery Optimizer** Aug 2026
*Python, XGBoost, TreeSHAP, SciPy* / *GitHub* / *Interactive Demo* / *Risk Modeling & Optimization*

- Engineered a dual-propensity classification framework across 20,000 delinquent accounts, decoupling behavioral **Willingness to Pay** from financial **Ability to Pay** to optimize call-center targeting and eliminate recovery budget waste.
- Audited and eliminated 9 look-ahead data-leakage variables; benchmarked models using global banking risk standards: **ROC-AUC 0.7226**, **Gini Coefficient 0.4451**, and **KS Statistic 34.15%**.
- Interpreted feature mechanics via **TreeSHAP**: proved Willingness is governed by promise-to-pay adherence and contact responsiveness, while Ability is driven by salary credit frequency and debt-to-income ratio.
- Formulated a 2×2 decision matrix under a 30% call-center headcount constraint, yielding an estimated **INR 36.97 Crore net cash recovery uplift** over the 34% random allocation baseline.

**RailCross — Transit Logistics & Uncertainty-Calibrated Predictor** Jul 2026
*Python, scikit-learn, TypeScript, Next.js* / *GitHub* / *Live App* / *Spatial ML & Selective Classification*

- Developed a spatial-temporal classification engine predicting railway level-crossing states and delay queues from traffic flow telemetry and stoppage duration signals, visualized on an interactive Next.js map interface.
- Enforced generalization by withholding 3 out of 14 crossings entirely as a spatial holdout, outperforming rule baselines with **ROC-AUC 0.888** and **F1 0.593** (vs. 0.16 rule baseline).
- Implemented an uncertainty-based selective abstention band that lifts prediction precision on accepted cases to **∼90%**, intentionally declining to predict when sensor signals are stale or ambiguous.
- Deployed model inference via a serverless microservice on Cloudflare Workers, achieving sub-100ms response times for real-time state classification queries.

## Education

**National Institute of Technology Jamshedpur** 2024 – 2028
*B.Tech, Production and Industrial Engineering* *Jamshedpur, India*

- **Relevant Coursework:** Probability & Engineering Statistics, Operations Research, Data Structures & Algorithms (DSA), Optimization Theory, Database Systems (DBMS).

## Achievements & Leadership

- **ET AI Hackathon 2026:** Engineered PRAHARI, a client-side fraud detection engine combining character-trigram Naive Bayes with a rule engine; selected among top standout AI projects.
- **Search Ranking Benchmark:** Trained pairwise XGBoost ranker on MSLR-WEB10K benchmark with 136 features (GitHub), achieving **NDCG@10 0.5581** and **MRR 0.8516**.
- **National Competitions:** Finalist participant at **XAVENIR 2026** (hallucination detection) and **MOVES 2026**.
- **Core Team Member, Phocus (NIT Jamshedpur):** Led photography, event media coverage, and technical equipment staging for annual college fests.